Shard Heist

The game rules, for both rooms. Built from `rules/*.md`.

Game rules

> These documents describe what the game does, as implemented in the code. When a rule changes, update the document, the code and the tests together, then run `npm run docs` to rebuild the printable version.

The game has two rooms that are played one after the other. Until the group connects them, each room is its own page.

Room	Document	Code with the rules	Tests
1 · Plot Twist: Cell Block D (escape room)	room-1-escape-room.md	`escape-room/src/rules.js`	`tests/escape-rules.test.js`
2 · Shard Heist (police chase)	room-2-shard-heist.md	`src/core.js` (rules), `src/game.js` (pickups, police AI)	`tests/rules.test.js`, `tests/story.test.js`, `tests/input.test.js`
Both rooms: every setting and option	settings.md	`src/core.js`, `escape-room/src/rules.js` (checks of stored values)	`tests/rules.test.js`, `tests/escape-rules.test.js`, `tests/browser.test.js`

The whole game in one paragraph

You were the detective on the Ashford necklace case. Now you are inmate 4471, convicted of stealing it. **Room 1:** you have ten minutes in a prison washroom to find the code for a hatch in the floor. If you get out (Ending A), a note shows that the guard wanted you out, and a car is waiting at the end of the tunnel. **Room 2:** the police chase begins. You drive a getaway vehicle through Lisbon's Baixa, on the highway and across the countryside, collecting star shards while up to four police cars and a helicopter hunt you.

One night, one look

- **Time:** both rooms are set on the same night: room 1 around 02:15, room 2 from about 02:30.
- **Design:** both rooms share one design system: the same background, glass panels, fonts, buttons and cards. Each room keeps its own accent colour: **orange** in the prison, **cyan** on the run. Hale's radio messages in room 2 are orange because he comes from room 1.

- A test (`tests/page.test.js`) checks that the shared design tokens and the story links between the rooms stay in place.

How the two rooms will connect (planned, not built)

- Ending A of room 1 shows a button that opens room 2.
- Ending B of room 1 stays in room 1 (try again).
- Open question for the group: does room 2 then start in free ride, or straight in the story mode *The Necklace Job*?

Words used in these documents

Word	Meaning
Accepted action	The player did something allowed. The state changes and is saved.
Refused action	The player tried something not allowed. Nothing changes, and the player is told why.
State	Everything the game remembers at one moment: where you are, items, time left, and so on.
Scene	A place you can stand in room 1 (the door, the sinks, ...).
Hotspot	Something you can use in a scene, shown as a pulsing marker or a "Walk to" button.
Tick	One step of the clock, many times per second.

Room 1 · Plot Twist: Cell Block D

> This document describes the rules in `escape-room/src/rules.js`. Values marked **(open)** are not decided yet.

Goal

Open the hatch in the floor of the toilet stall and climb out **before the clock reaches zero**.

Story

1. **Briefing.** You were the detective on the Ashford necklace case. Tonight you are inmate 4471, convicted of stealing the necklace: someone framed you.
2. Officer Hale walks you to washroom 2, locks the door and says: "Ten minutes. Not a second more." He waits outside.
3. Inmates whisper about a hatch in the floor of this washroom, locked with a four-digit code that is hidden somewhere in the room.

The clock only starts when the player presses **Start the clock** after the briefing.

Time

Rule	Value
Time limit	**10:00** (open: 10 or 15 minutes; `TIME_LIMIT`)
Warnings from Hale	at 5:00, 2:00, 1:00 and 0:30, each only once
Wrong code	costs **30 seconds** (`WRONG_CODE_PENALTY`), never below 0:00
Jammed door	when the clock hits 0:00 the first time, you get **one extra minute** (`WEDGE_GRACE`), once only
The clock stops	in the briefing, in the pause menu (which also opens when the browser tab is hidden or the window loses focus), and after an ending
The clock keeps running	while reading a clue and while turning the lock

Scenes and how to move

Scene	What is there	You can walk to
The Door (start)	Steel door with a barred window, Hale outside; mop in a bucket	Sinks, Stall
The Sinks	Two steel sinks, a polished metal mirror, a soap dish, a hot tap	Door, Stall

The Stall	Steel toilet, scratches on the wall, a crooked tile, the hatch in the floor	Door, Sinks, Hatch
The Hatch	The trapdoor with a four-wheel combination lock (you kneel down)	Stall
The Tunnel	Only reached through Ending A	–

Walking anywhere else is refused: "You cannot get there from here."

Items (you can carry two)

Item	Where	What it does
Spoon	Soap dish at the sinks	Needed to pry the crooked tile off the wall
Mop	Bucket by the door	Optional. Used at the door ("Jam the door with the mop") it is placed under the door handle and leaves your inventory. This buys the extra minute.

Clues and the code

Clue	How to find it	Text
Mirror	Turn on the hot tap first. Then the steam shows letters someone traced. Without steam you only see your face.	"UNSINKABLE?"
Graffiti	Read the scratches on the stall wall	"She was unsinkable. She sailed only once. The year opens the floor."
Newspaper scrap	Pry the tile off with the spoon, then read the paper behind it	"...the 'unsinkable' liner went down on her first voyage. 15 April 19▒2..."

- The code is **1912**, the year the Titanic sank. It is the **same in every game** (open: should it change between games?).
- Players who do not know the history can still solve it: "sailed only once" gives the missing digit 1.
- Every clue you read is kept in your clue notes (3 at most). Reading a clue again does not count twice.

The lock

- Four wheels, each showing 0–9. Up after 9 is 0, down after 0 is 9.
- You can only turn the wheels and try the code while kneeling at the hatch.
- **Try the code:**

- Right code: the hatch opens, Ending A. - Wrong code: "Clunk. The lock holds." Hale hears the noise, and you lose 30 seconds. The number of wrong codes is counted.

Endings

Ending	When	What happens
A · Out below	The right code before time runs out	You climb down the ladder. A note is taped to it: "Took you long enough. A car is waiting at the end of the tunnel. — H." The guard wanted you out. That is the plot twist that leads to room 2.
B · Back to the cell	The clock reaches 0:00 (after the extra minute, if the door was jammed)	The door bangs open and Hale and two officers take you back to cell 4471.

The earlier decision: jamming the door early gives one more minute at the very end. That can turn Ending B into Ending A.

After an ending, every action is refused except **Play again / Try again**, which starts a new game from the briefing.

Refused actions (nothing changes, the player is told why)

The player tries to...	Message (short)
do anything before the briefing is finished	"Read the briefing first."
walk where there is no way	"You cannot get there from here."
take an item that is not here, or twice	"There is no ... here." / "You already have the ..."
pry the tile without the spoon	"...your fingers cannot get behind it. Something thin and hard would do."
jam the door without the mop, or a second time	"You need something long and solid..." / "The door is already jammed."
read the scrap while the tile is still on	"The tile is still in place."
use the tap away from the sinks, the lock away from the hatch	"There is no tap here." / "Kneel down at the hatch to reach the lock."
turn a wheel that does not exist	"The lock only has four wheels."
do anything after an ending	"The story is over. Start a new game to play again."

Saving

- The game is saved in the browser after **every accepted action** and every few seconds while the clock runs.
- After a reload, the player can **Continue** (with the time that was left) or start a **New game**, which asks for confirmation first.
- A saved game is checked field by field when loaded. Anything that is not a real save is ignored, and a new game starts.

Controls

Input	Action
Drag the mouse or a finger, or the arrow keys	Look around
Click a pulsing marker or the object itself	Use it
"Walk to" buttons (bottom right)	Move to another scene
"Clues" button (bottom left)	Read your clue notes
`P` / `Esc`	Pause (Esc also closes a card)
`M`	Sound on or off
Volume slider	On the pause card, 0 to 100 %. It is kept for the next visit
`Enter`	Confirm the main button on a card

Room 2 · Shard Heist

> This document describes the rules in `src/core.js`, and the pickups and police in `src/game.js`.

Where room 2 starts

Room 2 begins where room 1 ends. You (inmate 4471, the framed detective) have climbed out of the tunnel under Cell Block D. It is about 02:30 at night. In a lock-up garage, Officer Hale's getaway vehicle is waiting. Every police car in town is now hunting the escaped inmate.

Officer Hale is a secret ally. He believes you were framed, but he cannot prove it. In the story mode he helps over the radio; his messages are shown in **orange**, the colour of room 1. Police dispatch messages are **red**.

Two ways to play

- **Free ride:** pick a map and a getaway vehicle, then escape.
- **Story mode,* The Necklace Job:* three acts on the three maps, in order, with a story and three endings.

Vehicles

Vehicle	Top speed	Acceleration	Handling	Damage taken	Ramming
Car (muscle coupe)	100 %	100 %	100 %	100 %	1.0
Motorcycle (superbike)	112 %	130 %	140 %	100 %	0.4
Truck (tractor unit)	78 %	70 %	72 %	**55 %**	2.4

Percentages are relative to the map's top speed and to the car. The menu shows the same values as bars (1–5) and as km/h and 0–80 km/h times, computed from these numbers.

Maps

Map	Goal	Time	Car top speed	Police top speed	Shards	Repair kits
Lisbon · Baixa	Collect **all 8 shards**, then drive through the golden gate that appears under the Arco da Rua Augusta	3:00	122 km/h	119 km/h	8	3
Highway	Reach the finish **4 km** ahead. Shards are optional: **+4 s** each	1:50	200 km/h	187 km/h	12	2

Countryside	Reach the village at the end of the road. Shards: **+4 s** each	2:30	158 km/h	148 km/h	10	3

- A shard is picked up within 3.8 m; a repair kit within 3.6 m, and only when you have damage.
- Lisbon has traffic lights, other cars and three yellow trams. The trams only run on their tracks, a stone track bed with two rails set into the road: a clockwise loop through Rua da Conceição, Rua da Prata, the north side of the Praça do Comércio and Rua Áurea. As on real Lisbon streets, cars and the player can drive over the tracks. The trams stop at red lights and behind other vehicles. On the highway traffic drives both ways. On country roads there is oncoming traffic, and driving off the road cuts your top speed to 55 %.
- Each map has an edge you cannot cross. When you near it the screen turns red and a "turn back" arrow appears. In Lisbon the outer streets are part of the map, so there this only happens when you head for the edge.

The city map: Lisbon's Baixa

The Baixa is the centre of Lisbon, rebuilt as a street grid after the 1755 earthquake. The map is a simplified version of it: the blocks are square, and the grid has 7 streets each way. The real Baixa has a few more, so some are left out. The real Baixa is also flat; the hills on this map are a game decision.

Direction	Streets, in their real order
West to east	Rua do Crucifixo, Rua Áurea, Rua dos Sapateiros, **Rua Augusta**, Rua dos Correeiros, Rua da Prata, Rua dos Fanqueiros
North to south	Rua de Santa Justa, Rua da Vitória, Rua de São Nicolau, Rua da Conceição, Rua do Comércio, Praça do Comércio (north side), Avenida Infante Dom Henrique

- In free ride, the panel at the top left shows the street or square you are on. In story mode it shows the act instead.
- You start on Rua Augusta, facing south towards the Arco da Rua Augusta.
- **Squares:** the Rossio and the Praça da Figueira in the north, the Praça do Comércio by the river in the south. The statue on the Praça do Comércio stands in the middle of the road, so traffic passes on both sides.
- **Landmarks you can see but not reach:** the river Tejo, the 25 de Abril bridge, Cristo Rei and the castle hill. The Santa Justa lift stands on the map, and you can crash into it.
- **Parks:** there are three, which the real Baixa does not have. The story needs them for the sealed letter.
- **Houses:** closed blocks of Pombaline houses, wall to wall, as the Baixa was rebuilt after 1755: a ground floor plus 3 to 5 floors (OpenStreetMap gives most buildings 5 levels), arched stone ground floors with old shop fronts, the tallest windows and the balconies on the first floor, limewash colours, and tile roofs with dormers. Tiled fronts are the exception, as they are in the real Baixa.
- **Real buildings in their place:** the National Theatre and Rossio station north of the Rossio, the yellow arcades and corner towers of the Praça do Comércio, and the Sé, seen at the end of Rua da Conceição. Their position is simplified, and they are modelled freely, not measured.

- It is always about 02:30 at night. The clock on the arch shows it.

Driving

Rule	Value
Top speed	map top speed × vehicle speed, **minus up to 40 %** at 100 % damage
Throttle	never pushes you above the top speed
Brake	slows you down, then reverses, up to 40 km/h backwards
No pedal	the car rolls to a stop
Handbrake (`Space`)	slows you quickly and turns sharper
Brake lights	on while you brake, in either direction, and with the handbrake while moving
Reversing lights	white, on while you roll backwards
Skid marks and tyre squeal	with the handbrake above about 15 km/h, when braking above about 30 km/h, and in very fast corners (above about 80 km/h, steering hard). They only mark the road, and they are cleared when a new run starts

Hills. All three maps have hills, up to about 20 % steep. The Arco, the river front and the two ends of the highway are level. Uphill your top speed falls by up to 16 % and gravity pulls you back, half as strongly as it pushes you downhill. Downhill your top speed rises by up to 22 % and you coast on. The truck feels slopes most, the motorcycle least. Brakes and rolling friction hold a stopped vehicle in place.

Smooth picture. If the game runs below 48 frames per second for 2 seconds, it lowers the resolution one step (85, 70, then 60 %). If that does not help, the step is taken back. After 20 seconds at 57 or more it goes back up, at most twice. The choice of Ultra or Balanced stays yours.

Damage

Rule	Value
Crash	impacts under 3 m/s (about 11 km/h) are free; above that, damage = (impact − 3) × 1.6 × the vehicle's damage factor. One crash counts once: for 0.35 s after contact, only a harder hit costs more
25 % damage	the paint shows dents and scratches, and thin grey smoke rises from the engine
50 % damage	warning: "Engine damaged: top speed reduced." Loose parts and a smashed light; the engine misfires; the smoke turns dark
75 % damage	hood up (car), screen gone (motorcycle), hanging bumper and cracked windscreen (truck); a thick black column of smoke; sparks at speed
88 % damage	the engine is on fire

100 % damage	**vehicle wrecked: game over**
Repair kit	removes 40 % damage, and the damage you see goes with it

Police

Rule	Value
First police car	after 4 seconds; in the countryside once you are about 170 m along the road
Wanted stars	1 to 5, shown under the clock. A new star every fifth of the time limit or of the way to the end (shards in the city, distance on the open roads), whichever is further along. Each wrecked police car counts as 4 % more. Stars never fall.
New police car	until the number for your stars is reached, see the table below
Ramming	every hard hit lowers a police car's health by impact × your ramming power × 0.09; a police car is wrecked when its health is gone and comes back after the time in the table below
Far behind	a police car more than 140 m away drives 20 % faster; one more than 230 m away in the city, or 320 m on the open roads, is sent back near you with full health
Hills	police slow down uphill and speed up downhill as you do; other traffic keeps its speed
Bust meter	fills while a police car is **within 9 m** and you are **slower than 4.5 m/s** (about 16 km/h): full in 2.2 s. Otherwise it drains (empty from full in 3 s).
Busted	bust meter at 100 %: game over

Stars	Police cars	Police speed	New car every	Wrecked car returns after	Health
★	1	100 %	3 s	9 s	3
★★	2	100 %	3 s	9 s	3
★★★	3	103 %	3 s	8 s	3
★★★★	3	106 %	2.5 s	6 s	3.5
★★★★★	4	110 %	2 s	4.5 s	4

Helicopter. After 60 seconds on the run (or sooner with the fifth star), a police helicopter joins the chase for the rest of the run and hunts you with its searchlight. Its searchlight follows you at 94 % of the police speed. While you are in the light (within 9 m of its centre), the police drive 6 % faster, and in the city they see you from 170 m instead of 110 m. It cannot bust you or damage you. An undamaged car or motorcycle at full speed on a clear, level road stays ahead of the light; the truck, and a badly damaged car, cannot.

Radio

Seven stations, or off: Reggae, Techno, Hardstyle, Schlager, Sing-Along Pop, Classical and Heavy Metal. Choose in the menu, or at any time with `R` or the radio button. The music plays while you drive, and is silent in the menu and in pause.

All of it is composed and synthesised by the game, with no music files. The tunes are generated by the game. The classical station plays Bach and Beethoven, whose works are in the public domain.

Ending a run

Result	When
Clean getaway	Lisbon: reach the gate under the arch. Highway: pass the 4 km mark. Countryside: reach the village
Busted	the bust meter is full
Wrecked	damage reaches 100 %
Time's up	the clock reaches 0:00

Stars and best times

- **Drive time.** In the city this is the time used. On the highway and in the countryside it is the time used plus the bonus seconds the shards gave you, so collecting shards never makes a run look slower.
- **Stars:**

Map	★★★	★★	★
City	more than 40 % of the time left	more than 20 % left	otherwise
Highway	under 88 s	under 105 s	otherwise
Countryside	under 115 s	under 135 s	otherwise

- These limits are for the car. For the other vehicles the drive time counts multiplied by their speed factor (motorcycle 1.12, truck 0.78), so each can earn three stars.
- More than 50 % damage costs one star, but never the last one.
- The best drive time per map is saved in the browser. Saved values under 10 s or over 15 minutes are ignored, because no run can take that long or that little.

Story mode: The Necklace Job

Act	Map	Goal	Special rule
1 · The Scatter	Lisbon	Recover the 8 shards, then the gate	An optional **sealed letter** is hidden in a park. It is **not shown on the radar**.

2 · The Courier	Highway	Catch the gold-marked courier lorry (get within 9 m) to take the **clasp**, then the finish	The **toll at the finish only opens with the clasp**: without it you stop there
3 · Homecoming	Countryside	Drive back to the Ashford mansion	The confrontation decides the ending

Ending	When
A · The truth	You found the letter: it proves Lady Ashford staged the theft. Hale gives you your badge back.
B · Back in 4471	No letter: Lady Ashford hands you to the police, and you go back to cell 4471.
C · Caught	Busted, wrecked or out of time in any act: back to Cell Block D. You can retry that act.

Retrying an act: a letter found during the failed attempt does **not** count. A letter found in an earlier, finished act does.

Radio messages (Hale in orange, police dispatch in red) appear at fixed times in each act, each only once.

Menu actions (accepted and refused)

Button	Allowed when
Start the getaway	in the main menu
Story mode	in the main menu, or after the final ending
Start act 1	on the story's opening card
Drive again / Restart	after a run, or in the pause menu
Resume	in the pause menu
Main menu	anywhere except while driving
Next act	after a finished act, but not after the last one
Retry act	after Ending C, or in the pause menu during a story

Any other button or action is refused, and no card button works while you are driving.

Saved settings

The last map and vehicle, the best times, the radio station, the graphics, the steering and both volumes are saved in the browser. Every saved value is checked when loaded; invalid values fall back to Lisbon, Car, radio off and the defaults. All settings are explained in Settings and options.

Controls

Input	Action
`W A S D` or arrow keys	Drive
`Space`	Handbrake
`P` / `Esc`	Pause
`M`	Sound on or off
Volume sliders	In the menu and on the pause card: `Game` (engine, effects, sirens) and `Music` (the radio), 0 to 100 %. They are kept for the next visit
`R`	Next radio station (`Shift` + `R` for the one before)
`Enter` / `Space`	Press the button in focus. With no button in focus, `Enter` presses the card's main button. While driving, `Space` is the handbrake
`Tab`	Move between the buttons of a card
Touch pads	Steer on the left; brake and gas on the right. Both together are the handbrake
Tilt (phones)	Hold the phone sideways and lower one side, like a steering wheel: from 3° it starts to steer, at 24° it is at full lock. The steering buttons go away while tilt works. Upright, or if the sensor is refused or missing, the buttons stay. Choose `Tilt` or `Buttons` in the menu or on the pause card. The sensor is only read in the browser and nothing is stored or sent

Settings and options

> Every setting the player can change, in both rooms: where it is, what each choice does, and what the game remembers. All of it is stored only in this browser (`localStorage`). Nothing is sent anywhere, and there are no accounts.

Room 2 · Shard Heist: the main menu

Setting	Choices	What it does
Map	Lisbon (Baixa), Highway, Country Roads	Where the run takes place. Each map has its own goal, time limit and top speed (see *Maps* in the room 2 rules)
Vehicle	Car, Motorcycle, Truck	Speed, acceleration, handling, how much damage a crash does and how the police can push you. The card under the choice shows the numbers
Mode	Start the getaway (free ride), or the story card *The Necklace Job*	Free ride is one run on the chosen map. The story plays all three maps in order, with Hale on the radio and three endings
Radio	Off, Reggae, Techno, Hardstyle, Schlager, Sing-Along Pop, Classical, Heavy Metal	The station that plays while you drive. `R` changes it at any time, `Shift` + `R` goes back one
Graphics	Ultra, Balanced	**Ultra:** sharper picture (up to twice the screen resolution), smoothed edges, glowing lights and the finest shadows (4096 × 4096). **Balanced:** a lighter picture without the glow and with smaller shadows (2048 × 2048). Computers start in Ultra, phones in Balanced
Steering (phones only)	Tilt, Buttons	**Tilt:** hold the phone sideways and lower one side like a steering wheel. **Buttons:** steer with the arrow pads. Tilt is the default on a phone with a motion sensor
Volume	`Game` 0–100 %, `Music` 0–100 %	`Game` is the engine, tyres, sirens, the helicopter and all effects. `Music` is the radio. The music plays for a moment while its slider moves, so it can be heard; letting go of the game slider plays a short tone

Room 2: during a run

Setting	How	What it does
Pause	`P` , `Esc` or the pause button	Stops everything: the clock, the police, the traffic and the music. The pause card shows the shards, time, damage and bust meter
Pause card	Resume, Restart (or Retry act in the story), Main menu	It also has the steering choice (phones) and both volume sliders

Sound	`M` or the speaker button	Turns all sound off or on, without changing the volume sliders
Radio	`R` or the radio button	Next station; the name is shown under the clock

Room 2: what the game sets by itself

Setting	Rule
Resolution	If the game runs below 48 frames per second for 2 seconds, the resolution drops one step: 85 %, then 70 %, then 60 %. A step that does not make the game faster by at least 5 % is taken back and the setting stays. After 20 seconds at 57 frames per second or more it goes one step back up, at most twice. Choosing Ultra or Balanced starts again at full resolution
Shadows	The area that gets shadows (120 m wide) follows the player in whole shadow pixels, so shadow edges do not crawl over the walls
Phones	Phones start in Balanced and use shadows of half the size
Less motion	If the system asks for reduced motion, the camera does not shake after crashes, the radar sweep stops and the menu turns more slowly
Hidden tab	When the tab is hidden or the window loses focus, the run pauses and the sound stops

Room 1 · Plot Twist: Cell Block D

Setting	How	What it does
Continue or New game	Shown after a reload, if there is a saved game	Continue goes on where you were, with the time that was left. New game asks first, then deletes the save
Pause	`P`, `Esc` or the pause button	Stops the clock. The pause card has Resume, Restart and the volume slider
Volume	Slider on the pause card, 0–100 %	All sounds of the washroom: the light's hum, dripping water, the tap and the effects. A short tone plays when it is let go
Sound	`M` or the speaker button	All sound off or on
Look around	Drag, or the arrow keys	Turns the view in the current scene

What is remembered, and what happens to bad values

Every stored value comes from outside the program, so each one is checked when it is loaded. A value that is missing or wrong is ignored and the default is used; the game never breaks because of it.

Stored	Room	Default if missing or invalid
Map and vehicle	2	Lisbon and the car
Best time per map	2	No best time (only 10 s to 900 s counts)
Radio station	2	Off

Graphics	2	Ultra on a computer, Balanced on a phone
Steering	2	Tilt
Game and music volume	2	100 % each; a value outside 0–100 % is cut to the range
Saved game	1	No saved game; any save that no real game could produce is refused
Volume	1	100 %

To reset everything, clear this site's data in the browser settings.